

Formula sheet

Equations	
Newton's law of universal gravitation	$F = G \frac{m_1 m_2}{r^2}$ where $G = 6.67 \times 10^{-11} \text{ Nm}^2\text{kg}^{-2}$
Weight and mass equation	$W = m \cdot g$ where g is the gravitational field strength On Earth, $g = 9.81 \text{ N kg}^{-1}$
Mole conversions	$m = n M$ Where m = mass, n = # of moles, and M = molar mass
Avogadro's constant	$6.02 \times 10^{23} \text{ mol}^{-1}$

The Periodic Table

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	1 H 1.01			Atomic number Element Relative atomic mass														2 He 4.00
2	3 Li 6.94	4 Be 9.01											5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18
3	11 Na 22.99	12 Mg 24.31											13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95
4	19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.87	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.38	31 Ga 69.72	32 Ge 72.63	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.90
5	37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.96	43 Tc (98)	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41	49 In 114.82	50 Sn 118.71	51 Sb 121.76	52 Te 127.60	53 I 126.90	54 Xe 131.29
6	55 Cs 132.91	56 Ba 137.33	57 † La 138.91	72 Hf 178.49	73 Ta 180.95	74 W 183.84	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59	81 Tl 204.38	82 Pb 207.2	83 Bi 208.98	84 Po (209)	85 At (210)	86 Rn (222)
7	87 Fr (223)	88 Ra (226)	89 ‡ Ac (227)	104 Rf (267)	105 Db (268)	106 Sg (269)	107 Bh (270)	108 Hs (269)	109 Mt (278)	110 Ds (281)	111 Rg (281)	112 Cn (285)	113 Nh (286)	114 Fl (289)	115 Mc (288)	116 Lv (293)	117 Ts (294)	118 Og (294)
†				58 Ce 140.12	59 Pr 140.91	60 Nd 144.24	61 Pm (145)	62 Sm 150.36	63 Eu 151.96	64 Gd 157.25	65 Tb 158.93	66 Dy 162.50	67 Ho 164.93	68 Er 167.26	69 Tm 168.93	70 Yb 173.05	71 Lu 174.97	
‡				90 Th 232.04	91 Pa 231.04	92 U 238.03	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (262)	

Section 1: Quickfire questions

1. Which is the strongest of the four fundamental forces?

2. Which has the largest range?

3. Which subatomic particles are found in the nucleus of an atom?

4. What is the unit of mass?

5. What is the unit of weight?

Section 2: Short Answer Questions

4. *Elemental Composition:*

- o An atom has 12 protons, 12 neutrons, and 10 electrons.
 - a. What is the charge of this atom?

b. What is the atomic number?

c. What is the mass number?

5. Describe what an isotope means:

6. Describe what an ion means:

7. What does the “relative” in relative atomic mass mean (Eg. Hydrogen mass is 1.01)?

8. *Difference Between Concepts:*

- o Describe the differences between **mass** and **weight**, and explain why astronauts weigh less on the Moon.

9. *Calculation of Weight:*

- o A dog has a mass of 25 kg. Calculate its weight on Earth, where $g=9.81$

Section 3: Periodic Table Practice

7. Fill in the table for the following elements:

Element Symbol	Atomic Number	Mass Number	Protons	Neutrons	Electrons
Mg	12	24			
F	9	19			

8. Draw the electron shell diagrams for the following:

a. Sodium (Na)

b. Fluorine (F)

Describe why atoms in the periodic table have a net charge of 0.

Section 4: Application and Analysis

9. A necklace contains 184 g of gold (Au)

a. Calculate the number of moles of gold in the necklace.

b. How many atoms of gold are in the necklace?

A student requires **2 moles of glucose**, which has the formula **C₆H₁₂O₆**, to carry out an experiment.

10. Calculate the **mass of C₆H₁₂O₆** that the student will need to weigh out for this reaction.

11. Calculate the **number of molecules** in 1.5 moles of C₆H₁₂O₆.

12. Calculate the **number of oxygen atoms** in 1.5 moles of C₆H₁₂O₆.

13. Two students have been separated in class and are sitting at a distance of 12m from each other. Their masses are 67kg and 54kg. Calculate the gravitational force between the two students.

14. Calculate the gravitational force a small car (400 kg) experiences on the surface of earth.

15. Between these three forces: the weak interaction, gravitational force and electromagnetic force, describe on what scale they are dominant and give an example for each.

Section 5: Scientific Inquiry

16. *Scenario:*

A scientist observes that the gravitational field strength on a mountain peak is slightly less than at sea level.

a. Suggest a reason for this difference.

17. Predict whether the gravitational field strength on Mars would be higher or lower than on Earth and justify your answer.
